

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

Dec 2023

Course Code: **EE101R01**

Course: **BASIC ELECTRONICS ENGINEERING**

QP No. : **UD039-1**

Duration: **3 hours**

Max. Marks: **100**

PART- A

Answer all the questions

10 x 2 = 20 Marks

1. Define knee voltage.
2. Identify and compare two optoelectronic devices, one which accepts light as input and other which emits light as output.
3. Draw the positive clamper circuit with input and output waveforms, consider input as square waveform.
4. The current gain of a CB transistor is 0.98. If the emitter current is 7 mA, find base current.
5. Sketch out the CE configuration with PNP transistor.
6. Give the doping level for transistor terminals.
7. Define drain resistance and amplification factor of FET.
8. The gate cut-off voltage of the N-channel JFET is given by $V_{GS(OFF)} = -6\text{ V}$ and $I_{DSS} = 12\text{ mA}$. Determine the value of I_D when $V_{GS} = 0\text{ V}$ and -2 V .
9. How will you measure current using CRO?

10. Find the value of multiplier resistance on a 10V range dc voltmeter which uses a full-scale deflection 1mA meter movement with an internal resistance of 100Ω .

PART- B

Answer any FOUR questions

4 x 15 = 60 Marks

11. A 230V, 50Hz, voltage is given to a primary winding of 5:1 step down transformer in a bridge rectifier circuit. The load resistance is 400Ω . If the diode forward resistance is 10Ω and the resistance of transformer secondary is 30Ω , find peak value of voltage, peak value of current, input power, output power, efficiency, form factor, peak factor and ripple frequency.
12. For the sinusoidal input, construct and explain the following clippers.
a) Series Positive Clipper b) Shunt Negative Clipper c) Series negative clipper with positive reference.
13. a) Find R_B & R_C for a fixed bias circuit with an operating point of $V_{CE} = 7\text{ V}$ and $I_C = 3\text{ mA}$. Let $V_{CC} = 15\text{ V}$, $\beta = 100$ and $V_{BE} = 0.6\text{ V}$ and also draw the circuit. (7)
b) The Silicon transistor in the fixed bias circuit has $\beta = 100$. Draw the DC load line and find the operating point if $R_B = 300\text{ k}\Omega$, $R_C = 2\text{ k}\Omega$ and $V_{CC} = 10\text{ V}$. (8)
14. Identify the transistor configuration, which one is used for high frequency circuits and explain the input and output characteristics with necessary diagrams using NPN transistor.
15. Elucidate the construction and working of n-channel D-MOSFET with its characteristics curves.

16. a) The coil of a PMMC instrument has 200 turns, on a former that is 18 mm wide, the effective length of the conductor being 25 mm. It moves in a uniform field of flux density 0.2 Tesla. The deflecting torque is 40×10^{-6} Nm. Calculate the current through the coil. (3)
- b) With the diagrammatic illustration explain the construction and working of Cathode Ray Tube in detail. (12)

PART- C

Answer the following

1 x 20 = 20 Marks

17. a) Sketch the regulator block diagram and explain. Also analyze the Zener diode regulator circuit for the conditions (i) when input is varying with constant load and (ii) for varying load with constant input. (15)
- b) Justify the term 'CMOS as an inverter'. (5)
